Question 3 — A Complete Report

Cleaning, Principal Component Analysis, and Clustering of a High-Dimensional Spectroscopic Dataset

SIMC 2026 Endeavour Challenge (Parts a–h)

May 2026

Abstract

This report works through the whole of Question 3 in detail and from the ground up. We are given a table of 1000 infrared spectra, each recorded at 3401 frequencies, and asked to clean it, analyse it, and ultimately recover how many distinct food samples it contains. Along the way we meet — and carefully define — the row norm, mean-centering, variance, the covariance and Gram matrices, eigenvalues and eigenvectors, Principal Component Analysis (PCA), the singular value decomposition, cosine similarity, and clustering. The headline results are: 40 blank vials and 36 anomalous high-norm vials are removed (leaving 924 clean spectra); the data is strongly low-dimensional, with a dominance ratio $\lambda_1/\lambda_2 \approx 3.53$ and a clear "elbow" at $k^\star = 8$ signal directions; the leading principal direction is an interpretable contrast between molecular bands; and clustering — carried out two independent ways that we prove must agree — shows the genuine vials fall into well-separated groups of about 37 repeats each, giving a final count of **26 unique food samples**.

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1 Introduction and roadmap

Imagine a blind taste test. A thousand small vials are filled with fermented, caffeinated drinks — kombucha, cocoa drinks, coffee liqueurs, Chinese tea — and an instrument shines infrared light of many different frequencies through each one, recording how strongly the liquid absorbs at each frequency. The result is a long list of numbers per vial, a *spectrum*, which acts as a chemical fingerprint. The labels saying which vial held which drink have been lost, and our job is to use linear algebra to reconstruct the structure of the experiment: which measurements are corrupt, which frequencies carry information, what the dominant patterns of variation are, and — finally — how many genuinely different drinks were tested.

The mathematics needed is introduced as we go, so no prior linear algebra is assumed. The report is organised as follows. Section 2 defines every tool we will use. Section 3 describes the dataset and a subtlety about how it is stored. Then each part of Question 3 has its own section: cleaning (§4), variance and the covariance matrix (§5), the eigenspectrum and PCA (§6), the leading eigenvector (§7), clustering in two different spaces (§8–9), the comparison of those two approaches (§10), and the final count of unique samples (§11). Section 12 ties the threads together, and the appendices give a glossary and a table of all key numbers.

2 Mathematical background, from scratch

2.1 Vectors and the norm

A **vector** is just an ordered list of numbers, written as a column, $x = (x_1, x_2, \ldots, x_M)$. Each number is a *component*. For us a single spectrum is a vector of $M = 3401$ absorbance values. The **(Euclidean) norm** of a vector measures its overall size,

$$\|x\| \;=\; \sqrt{x_1^2 + x_2^2 + \cdots + x_M^2}.$$

Geometrically it is the length of the arrow from the origin to the point x. Because every component is squared before being summed, a few large values dominate the norm — a fact we exploit in Section 4.

2.2 Matrices, transpose, and products

A **matrix** is a rectangular grid of numbers. The **transpose** $A^\top$ is obtained by swapping rows and columns, so $(A^\top)_{ij} = A_{ji}$; a $p \times q$ matrix becomes $q \times p$. The **dot product** of two vectors, $a \cdot b = \sum_i a_i b_i$, multiplies matching components and adds them; it is large and positive when the two vectors point the same way, zero when they are perpendicular. Matrix multiplication generalises this: entry (i, k) of AB is the dot product of row i of A with column k of B.

2.3 The design matrix

Our data is a **design matrix** $X \in \mathbb{R}^{N\times M}$ whose rows are **samples** (one vial each) and whose columns are **features** (one frequency each); the entry X_{ij} is the absorbance of sample i at frequency j. Here $N = 1000$ and $M = 3401$.

2.4 Mean, variance, and mean-centering

The **mean** of a feature is its average over the samples, $\bar{X}_j = \frac{1}{N}\sum_i X_{ij}$. **Mean-centering** subtracts this average from every entry of the column, producing $\widetilde{X}_{ij} = X_{ij} - \bar{X}_j$; afterwards every feature averages zero. The **variance** of a feature is the average squared distance from its mean, $\mathrm{Var}_j = \frac{1}{N}\sum_i (X_{ij} - \bar{X}_j)^2$. A large variance means the feature swings widely between vials and is therefore informative; a near-zero variance means it is almost constant and tells us little.

2.5 The covariance matrix

The **covariance** of two features asks whether they move together. Collected for all pairs of the (mean-centred) features, the covariances form the **covariance matrix**

$$C \;=\; \tfrac{1}{N}\,\widetilde{X}^\top \widetilde{X} \;\in\; \mathbb{R}^{M\times M}, \qquad C_{ij} = \tfrac{1}{N}\sum_k \widetilde{X}_{ki}\widetilde{X}_{kj}.$$

An off-diagonal entry is positive when two features rise and fall together, negative when one rises as the other falls. The diagonal entry C_{jj} is the covariance of a feature with itself, which is exactly its variance. Crucially, C is *symmetric*: $C^\top = C$.

2.6 Eigenvalues and eigenvectors

For a square matrix C, a non-zero vector v is an **eigenvector** with **eigenvalue** λ if

$$Cv = \lambda v,$$

meaning that multiplying by C only *stretches* v (by the factor λ) without rotating it. Eigenvectors are the special, "natural" directions of a matrix.

2.7 The spectral theorem and positive semi-definiteness

Two facts about our C make everything well behaved. By the **spectral theorem**, a real symmetric matrix has only *real* eigenvalues and a complete set of mutually perpendicular (**orthonormal**) eigenvectors — a clean coordinate system. And C is **positive semi-definite (PSD)**: for every vector w, $w^\top Cw \geq 0$ (because, as we show later, it equals a sum of squares). A PSD matrix has all eigenvalues ≥ 0, so each eigenvalue is a genuine, non-negative variance.

2.8 Principal Component Analysis (PCA) in one paragraph

PCA is simply the eigendecomposition of the covariance matrix. Its eigenvectors are the *principal directions* of the data — the axes along which the samples vary most — and the corresponding eigenvalues say how much variance lies along each. Keeping only the few largest directions compresses the data while preserving almost all of its structure.

2.9 The singular value decomposition and the Gram matrix

Every matrix factors as $\widetilde{X} = U\Sigma V^\top$ (its **singular value decomposition**, SVD), where U and V have orthonormal columns and Σ is diagonal with the non-negative *singular values* σ_i. The columns of V are the eigenvectors of $\widetilde{X}^\top\widetilde{X}$ (feature space) and the columns of U are the eigenvectors of the **Gram matrix** $G = \frac{1}{\widetilde{N}}\widetilde{X}\widetilde{X}^\top \in \mathbb{R}^{\widetilde{N}\times\widetilde{N}}$ (sample space); both have eigenvalues $\sigma_i^2/\widetilde{N}$. The Gram entry G_{ij} is the similarity (dot product) between samples i and j. This shared-eigenvalue link (Question 2d) is the reason the two clustering routes in §8–9 must agree.

2.10 Cosine similarity

To compare the *shape* of two vectors regardless of their length we use **cosine similarity**,

$$\cos(a, b) = \frac{a \cdot b}{\|a\|\,\|b\|} \in [-1, 1],$$

the cosine of the angle between them: $+1$ for identical direction, 0 for perpendicular, -1 for opposite.

2.11 Clustering

A **cluster** is a group of points that lie close together and are well separated from other groups. We count clusters with two simple, library-free methods: *leader clustering* (walk through the points, opening a new cluster centre whenever a point is farther than a radius r from all existing centres) and *single-linkage connected components* (join two points if they are within a distance ε, and count the connected groups). When clusters are well separated, both give the same count over a wide range of r or ε.

3 The dataset and its orientation

The data is supplied as the file `mystery_matrix1.npy`. Loading it gives an array of shape $(3401, 1000)$.

Remark 3.1 (Two subtleties worth stating up front). First, the problem text quotes $M = 3041$ features, but the file genuinely has 3401 columns — the same digits transposed. We trust the data and use 3401. Second, the array is stored *transposed* relative to the "samples-as-rows" convention: its length-1000 axis is the samples. We therefore transpose it once, so that each row is one vial and each column one frequency, and we identify the sample axis as the one of length 1000 (matching the 1000 vials of the experiment). All values are non-negative and lie in $[0,\ 0.85]$, with no NaN/infinite entries.

4 Part (a): cleaning the data

Real measurements contain mistakes, and a sound analysis removes them first. Two distinct kinds of corrupted vial are present.

4.1 The first error: empty vials

A vial with no sample in it produces only faint background noise — a nearly flat spectrum of small values. To detect these without inspecting all 3401 numbers per vial, we reduce each spectrum to a single number, its row norm $\|x_i\|$. A genuine spectrum has tall absorption peaks and hence a large norm; an empty vial carries only noise and hence a small norm.

A data-driven threshold. Sorting the 1000 norms reveals a clean gap (Figure 1): about 40 vials sit at $\|x\| \approx 1.6$, then there is a void until the genuine vials begin at $\|x\| \approx 4.3$. Rather than hard-coding a cut-off, we place the threshold in the *middle of the largest gap*. This is robust — a small change in the data does not move the decision — and leaves no arbitrary constant to justify. Exactly $T = 40$ vials fall below it; they turn out to be the last 40 rows of the file, a hint that the corrupt rows were appended deliberately.

4.2 The other error: high-norm outliers

The problem warns of a second error. The *upper* tail of the sorted norms holds another clearly detached group: 36 vials at $\|x\| \approx 8.9$, separated from the bulk by a gap about five times wider than any internal gap and forming the contiguous block of rows 924–959. We flag these as the second error. We are careful to note that such a coherent, high-magnitude group could instead be a genuinely strong (concentrated) drink rather than a mistake; we therefore (i) provide independent evidence below that they form one tight, isolated cluster, and (ii) keep their removal reversible. (This judgement is revisited in §11, where it matters for the final count.)

4.3 Selection by a Boolean mask

We build two Boolean (true/false) masks — *empty* and *high* — and keep a vial only if it is in neither, i.e. *keep* $= \neg\,$empty $\wedge \neg\,$high. We then *index* the data with this mask rather than deleting rows. This is (1) non-destructive — the original matrix is never altered, so any decision can be undone; (2) composable — two independent rules combine in a single step; and (3) index-preserving — vial 193 is still vial 193 afterwards. The cleaned matrix has $\widetilde{N} = 924$ rows.

4.4 Validation

Three independent checks confirm the removal. *Visual:* removed spectra are flat noise while the kept spectra show sharp peaks (Figure 1). *Statistical:* the empty vials are 50.2% exact zeros against 24.5% for genuine spectra. *Structural:* in the reduced picture of §6, the high-norm group forms one isolated cluster whose removal lowers the cluster count by exactly one.

Remark 4.1 (Why 50.2% zeros for blanks but only 24.5% for real spectra?). Both numbers follow from a single mechanism: **clipping**. A detector cannot report a negative absorbance, so any reading that would come out below zero is recorded as exactly 0.0 — the signal is *rectified* at zero. The two fractions differ because the two kinds of vial spend different amounts of the frequency axis near zero.

Empty vials: 50.2%. A blank contains no sample, so at every one of its 3401 frequencies the detector sees only small electronic/thermal noise. That noise is symmetric about zero: at each frequency it is equally likely to land just above or just below 0, so its true mean is 0. Clipping converts every would-be-negative reading into an exact zero, and a zero-mean symmetric quantity is negative about half the time. Hence almost exactly half of a blank's readings — measured at 50.2% — are exact zeros, while the other half keep their small positive noise value. (This is why the figure is so close to a clean 50%: it is just "how often is the noise negative?")

Genuine spectra: 24.5%. A real drink absorbs strongly in the molecular bands — the spiky regions of Figure 2 — which lift those readings well above zero, so there the noise never makes them negative and they are essentially never clipped. Only in the flat *baseline* stretches *between* the peaks does the true signal sit near zero; there the noise is again clipped to exact zeros, exactly as in a blank. Because a real spectrum is baseline over only part of the axis (roughly a quarter of it), its exact-zero fraction is about 24.5% — the same $\approx$ 50%-of-the-baseline effect, but now applied to only the baseline portion rather than the whole spectrum.

So the contrast 50.2% vs. 24.5% is a direct physical fingerprint: a blank is "all baseline" (half its channels clipped), whereas a real drink is baseline only between its peaks (only a quarter clipped). This is an independent confirmation of the row-norm test — a vial flagged as empty by its small norm is also exactly the one whose readings are $\approx$ 50% zeros.

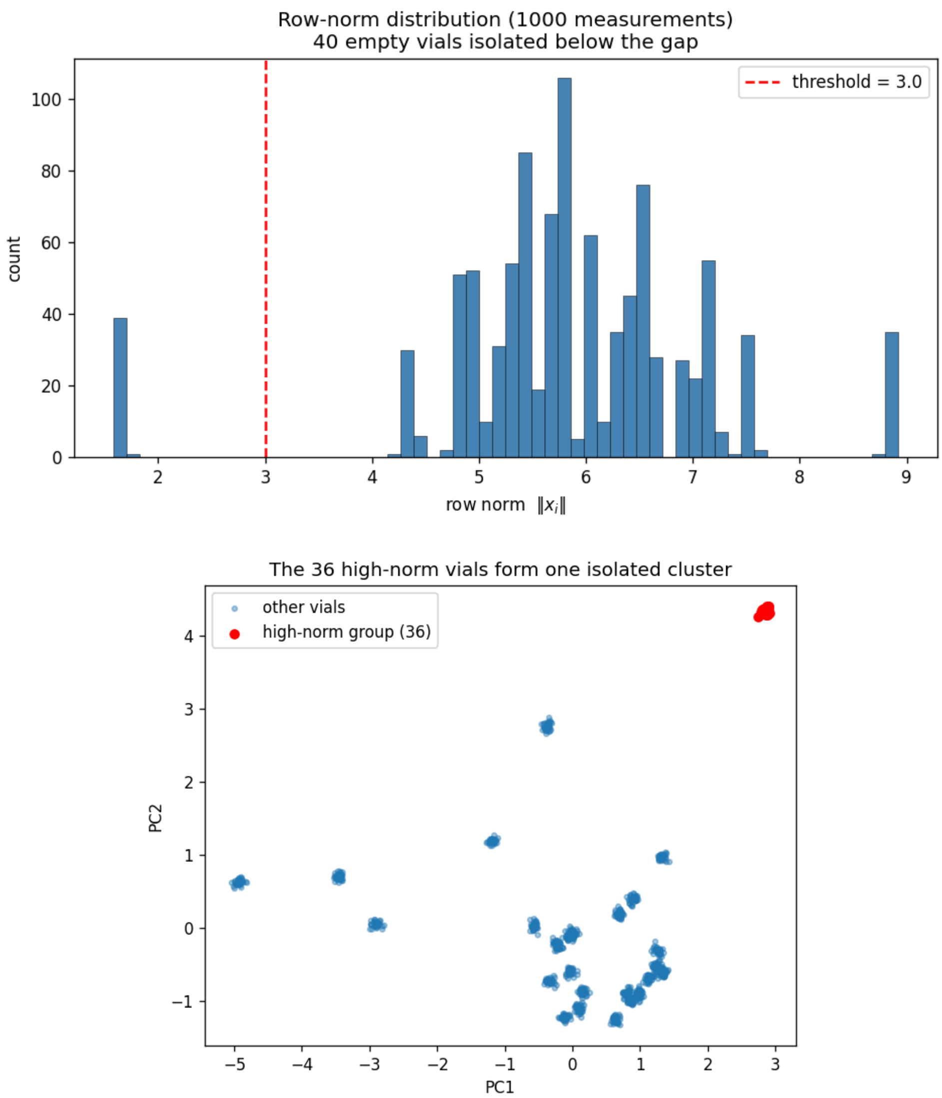

Figure 1: *Top:* the row-norm distribution; 40 empty vials sit below the gap (threshold marked). *Bottom:* the 36 high-norm vials form one tight, isolated cluster in the leading principal-component plane.

5 Part (b): variance and the covariance matrix

With clean data we ask which of the 3401 frequencies actually distinguish the samples.

We first **mean-center**, subtracting each feature's average so that the analysis describes how samples *differ* rather than the large average spectrum they all share. (Intuition: in a class where everyone is about 140 cm tall, what is informative is who is taller or shorter than average, not the 140 everyone shares.) Call the result $\widetilde{X}$.

Proposition 5.1. *$C = \frac{1}{\widetilde{N}}\widetilde{X}^{\mathsf{T}}\widetilde{X}$ is symmetric, and its diagonal entries are the per-feature variances: $C_{jj} = \frac{1}{\widetilde{N}}\sum_k \widetilde{X}_{kj}^2 = \mathrm{Var}_j$.*

Proof. $C^{\mathsf{T}} = \frac{1}{\widetilde{N}}(\widetilde{X}^{\mathsf{T}}\widetilde{X})^{\mathsf{T}} = \frac{1}{\widetilde{N}}\widetilde{X}^{\mathsf{T}}\widetilde{X} = C$. Setting $i = j$ in the definition gives $C_{jj} = \frac{1}{\widetilde{N}}\sum_k \widetilde{X}_{kj}^2$, which, as the columns are mean-centred, is precisely the variance of feature j. □

A practical consequence: to answer part (b) we need only the diagonal of C, so we compute the variances directly (one operation) and never form the full 3401×3401 matrix. We divide by $\widetilde{N}$ (the population variance, NumPy's default), which matches the definition of C.

Proposition 5.2 (Total variance is conserved)**.** *The **trace** $\mathrm{tr}(C) = \sum_j C_{jj}$ is the total variance and is invariant under any orthogonal change of basis $C \mapsto Q^{\mathsf{T}}CQ$; in particular $\mathrm{tr}(C) = \sum_i \lambda_i$.*

Proof. $\mathrm{tr}(Q^{\mathsf{T}}CQ) = \mathrm{tr}(CQQ^{\mathsf{T}}) = \mathrm{tr}(C)$ by cyclicity of the trace and $QQ^{\mathsf{T}} = I$. Diagonalising $C = Q\Lambda Q^{\mathsf{T}}$ gives $\mathrm{tr}(C) = \mathrm{tr}(\Lambda) = \sum_i \lambda_i$. □

Empirically $\mathrm{tr}(C) \approx 7.90$, which we will re-check against the eigenvalues in §6. Figure 2 plots C_{jj} against the frequency index: the variance is sharply concentrated in a few narrow bands — the molecular absorption regions where vials differ most — while most frequencies are flat. The single most variable frequency is $j^\star = 910$, with $C_{j^\star j^\star} \approx 0.0199$. The lesson is that only a handful of frequencies carry information: the data is effectively *low-dimensional*, exactly the structure PCA exploits next.

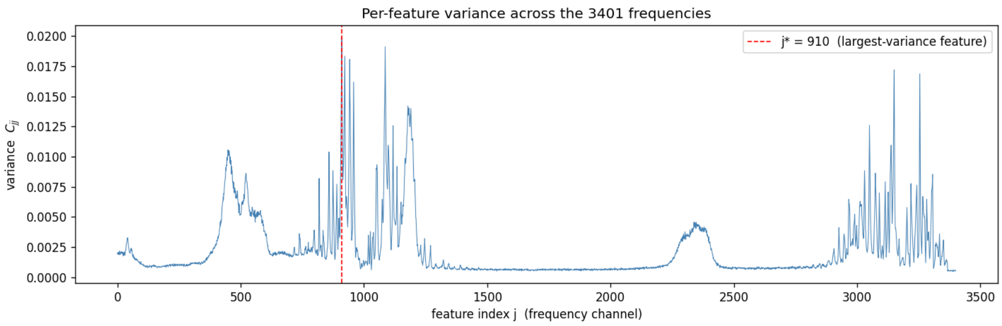

Figure 2: Per-feature variance C_{jj}. Variance concentrates in narrow molecular bands; the red line marks the maximal-variance feature $j^\star = 910$.

6 Part (c): the eigenspectrum (PCA)

We now diagonalise C. The eigenvectors are the principal directions; the eigenvalues rank them by importance.

Proposition 6.1 (Eigenvalues are variances; C is PSD)**.** *For a unit vector w, the variance of the data projected onto w is $w^{\mathsf{T}}Cw$. If $C\nu = \lambda\nu$ with $\|\nu\| = 1$ then $\nu^{\mathsf{T}}C\nu = \lambda$. Moreover $w^{\mathsf{T}}Cw \geq 0$ for all w, so every $\lambda_i \geq 0$.*

Proof. The projection of sample $\widetilde{x}_i$ onto w is the scalar $z_i = \widetilde{x}_i \cdot w$, with mean zero; its variance is $\frac{1}{\widetilde{N}}\sum_i z_i^2 = \frac{1}{\widetilde{N}}w^{\mathsf{T}}\widetilde{X}^{\mathsf{T}}\widetilde{X}w = w^{\mathsf{T}}Cw$. Substituting an eigenvector gives $\nu^{\mathsf{T}}C\nu = \lambda\|\nu\|^2 = \lambda$. Finally $w^{\mathsf{T}}Cw = \frac{1}{\widetilde{N}}\|\widetilde{X}w\|^2 \geq 0$, so $\lambda = \nu^{\mathsf{T}}C\nu \geq 0$. □

By the spectral theorem C has real eigenvalues and an orthonormal eigenbasis. We compute the decomposition with `numpy.linalg.eigh`, the routine specialised for symmetric matrices: it guarantees real eigenvalues and orthonormal eigenvectors and is faster and more stable than the general `eig`. We sort the eigenvalues in decreasing order and clip rounding-level negatives (of order 10^{-17}) up to zero, which is legitimate because C is PSD (Prop. 6.1).

6.1 The dominance ratio and the elbow

Table 1 lists the leading eigenvalues. The **dominance ratio** is $\lambda_1/\lambda_2 \approx 3.53$: one direction captures far more variance than any other, so the data has a strong principal axis. (Equivalently the *spectral gap* $\lambda_1 - \lambda_2$ is wide, which is exactly the condition that makes the Power Method of Question 2 converge quickly.)

We locate the **elbow** $k^\star$ — the boundary between meaningful "signal" eigenvalues and the flat "noise" floor — as the largest *multiplicative* drop between consecutive eigenvalues. (We use a ratio, not a difference, because the signal/noise transition is multiplicative on the logarithmic scale of Figure 3; a difference would simply track the huge λ_1.) The largest drop is $\lambda_8/\lambda_9 \approx 4.1$ (Table 1), giving $\boxed{k^\star = 8}$. It is not 7, because λ_8 still sits about $4\times$ above the floor (genuine signal); it is not 9, because λ_9 already lies on the flat floor (noise). Pleasingly, 8 is close to the $\sim$ 10 dominant molecules named in the problem: a mixture of a few chemical components spans only a few independent directions.

k	λ_k	λ_k/λ_{k+1}	variance %	cumulative %
1	2.4918	3.53	31.5	31.5
2	0.7052	1.20	8.9	40.5
3	0.5899	2.33	7.5	47.9
4	0.2535	2.06	3.2	51.1
5	0.1232	1.18	1.6	52.7
6	0.1045	1.35	1.3	54.0
7	0.0775	2.01	1.0	55.0
8	**0.0386**	**4.13**	0.5	55.5
9	0.0093	1.00	0.1	55.6

Table 1: The leading eigenvalues, the consecutive ratio (whose largest value, 4.13, marks the elbow at $k^\star = 8$), and the per-component and cumulative share of total variance. The full table for $k = 1 \dots 9$ is also saved as a loadable CSV alongside the notebook.

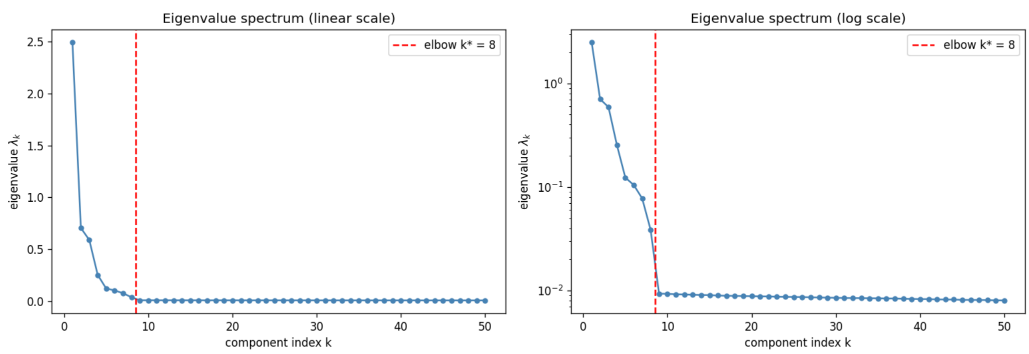

Figure 3: The eigenvalue spectrum on linear (left) and logarithmic (right) scales. The log plot shows a sharp elbow at $k^\star = 8$, after which the eigenvalues flatten onto a noise floor.

6.2 Cumulative variance: a cautionary tale

A popular rule keeps enough components to explain 95% of the variance. Here that rule is badly misleading. The 8 signal components explain only $\approx 55\%$ of the total variance, and reaching 95% would require $k = 667$ components (Figure 4). The reason is that the tiny noise floor, though negligible per component, is spread over $\approx$ 900 dimensions (the rank of $\widetilde{X}$ is at most $\widetilde{N} - 1$), so its contributions accumulate. The honest measure of dimensionality is therefore the elbow (8), not the 95% threshold (667). As a final check, $\sum_i \lambda_i \approx \mathrm{tr}(C) \approx 7.90$, confirming Prop. 5.2.

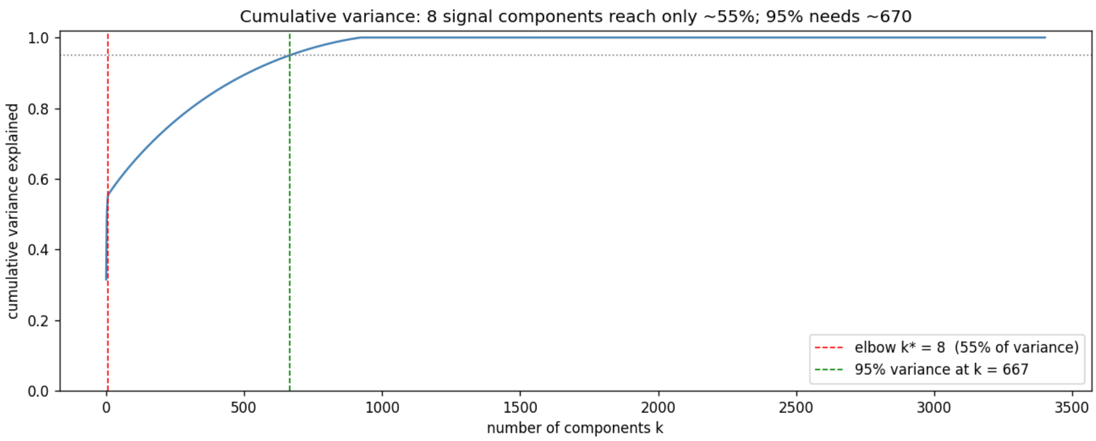

Figure 4: Cumulative variance versus the number of components. The elbow (red) reaches only $\approx$ 55%; the 95% mark (green) is not attained until $k = 667$.

7 Part (d): the leading eigenvector

The leading eigenvector $\nu_1 \in \mathbb{R}^M$ is the single direction of greatest variation. It lives in *feature space* and assigns a weight (a **loading**) to each frequency. Its practical importance: among all directions, the one-number summary $z_i = \widetilde{x}_i \cdot \nu_1$ (the **score**) is the most information-preserving way to describe a vial, because its scores have the largest possible variance (Prop. 6.1).

Two technical points. *Loadings vs. scores:* the loadings are the entries of ν_1 (how each frequency contributes to the pattern), while the scores place each sample along it. *Sign:* an eigenvector is defined only up to an overall sign (if $C\nu = \lambda\nu$ then $C(-\nu) = \lambda(-\nu)$), so we flip ν_1 to make its largest loading positive; this changes nothing mathematically.

Figure 5 (top) shows that ν_1 is supported on two narrow bands ($j \approx 900$–960 and $j \approx 3150$–3260) and takes *both signs* (about 27% of its loadings are negative). A mixed-sign direction is a **contrast**: it separates vials by the *balance* of absorption between two groups of frequencies, not by overall brightness.

Does ν_1 resemble a real measurement? Using cosine similarity, it matches the *centred* spectrum of vial 193 at cos ≈ 0.92, but only 0.67 against that vial's *raw* spectrum and ≈ 0.03 against the mean spectrum (Figure 5, bottom). The explanation is exactly the mean-centering of part (b): ν_1 lives in the centred space and is essentially orthogonal to the average spectrum, which dominates the raw spectra. So ν_1 is a *difference-spectrum* — the way the most-deviated concoction differs from the average — mirroring Question 1, where the dominant eigenvector was the system's long-time steady state.

8 Part (e): clustering in covariance (feature) space

We now compress each vial to a handful of coordinates and look for groups. With $V_4 = [\nu_1\, \nu_2\, \nu_3\, \nu_4]$ (the top four eigenvectors of C), the compressed coordinates are $Z = \widetilde{X} V_4 \in \mathbb{R}^{\widetilde{N} \times 4}$: row z_i holds the four scores of vial i. This is **dimensionality reduction** from 3401 numbers to 4, retaining the dominant variation. We

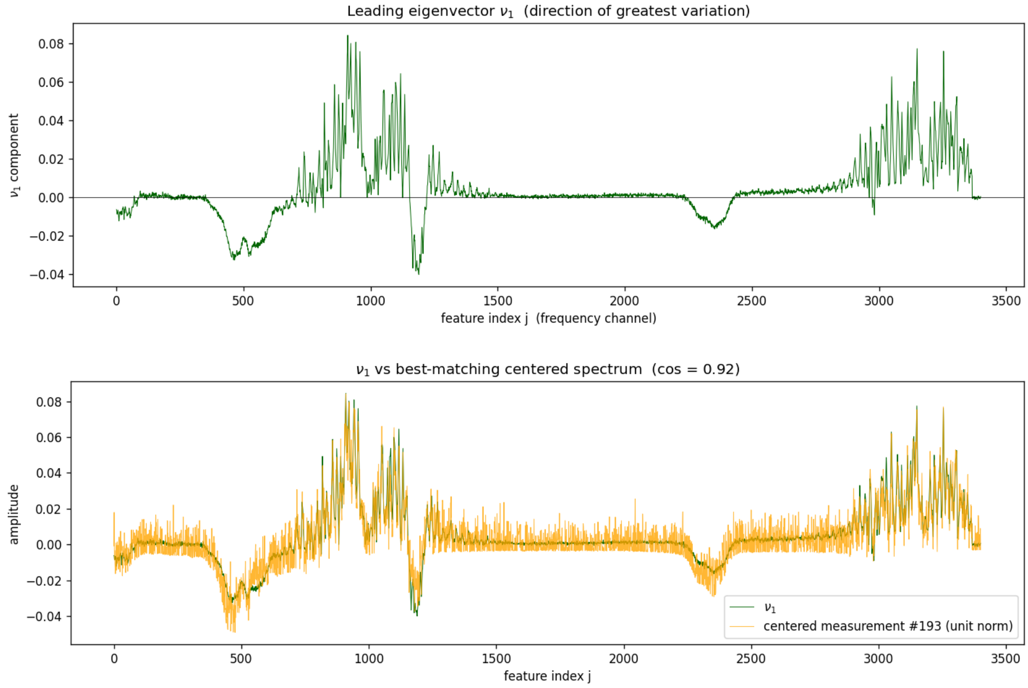

Figure 5: *Top:* loadings of ν_1, concentrated on two molecular bands with mixed signs (a contrast direction). *Bottom:* ν_1 overlaid on the best-matching centred spectrum (cos ≈ 0.92).

choose four because they lie within the eight signal directions of §6 yet are few enough to inspect as the $\binom{4}{2} = 6$ pairwise scatter plots (Figure 6), drawn with small, semi-transparent points so dense groups stay visible.

The panels show many tight, well-separated blobs. Counting them by leader clustering (radius a few multiples of the median nearest-neighbour spacing) gives 25 clusters, and the count is stable for radii from 6 to 12 times that spacing — so it is a property of the data, not of the threshold. Each blob is a set of vials with nearly identical spectra, i.e. one concoction.

9 Part (f): clustering in Gram (sample) space

The **Gram matrix** $G = \frac{1}{\widetilde{N}} \widetilde{X} \widetilde{X}^{\mathsf{T}}$ is the $\widetilde{N} \times \widetilde{N}$ companion of C; its entry G_{ij} is the similarity between vials i and j. Diagonalising G confirms it shares the nonzero eigenvalues of C exactly (the top eight match to all printed digits), as Question 2d predicts. Its eigenvectors are already $\widetilde{N}$-dimensional, so *each component is directly a vial's coordinate* — no projection of $\widetilde{X}$ is required. Scaling the eigenvectors by $\sqrt{\widetilde{N}\lambda} = \sigma$ puts them in the same units as the covariance-space scores of §8.

The same six scatter plots (Figure 7) and the same leader-clustering rule again give 25 clusters, stable over the same range of radii. The two scatter pictures reproduce one another up to per-axis sign flips, because eigenvectors are defined only up to a sign. We make this precise in the next section.

10 Part (g): comparing the two approaches

10.1 Do the two methods give the same number of clusters? Yes — necessarily.

Both routes return 25 clusters, and this is forced by the algebra. The mathematical link: if $C\nu = \lambda\nu$ then $\widetilde{X}\nu$ is an eigenvector of G with the same eigenvalue λ, so C and G share their nonzero eigenvalues. More strongly, with the SVD $\widetilde{X} = U\Sigma V^{\mathsf{T}}$ the covariance-space scores are $Z = \widetilde{X}V = U\Sigma$, while the (scaled)

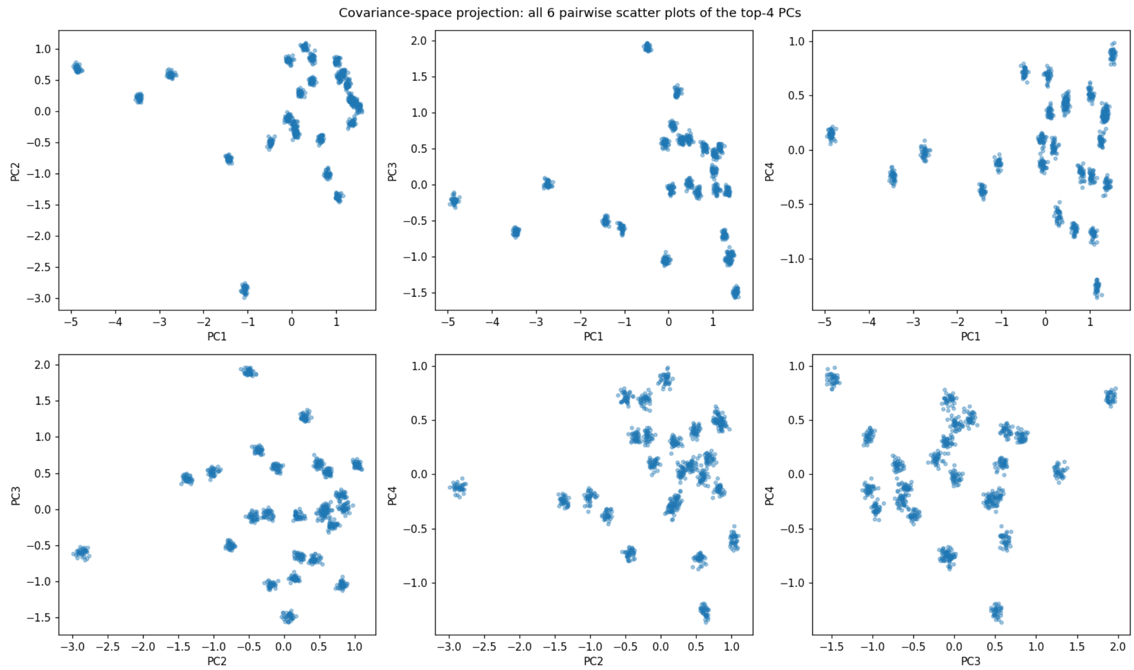

Figure 6: The six pairwise scatter plots of the top-4 covariance-space scores; each point is one vial. The data resolves into ≈ 25 well-separated clusters.

Gram coordinates are $Z_g = U\sqrt{\widetilde{N}\lambda} = U\Sigma$. **Both coordinate sets are the same matrix** $U\Sigma$, identical up to the arbitrary sign of each eigenvector.

We verify this directly by correlating the two coordinate sets component by component (Figure 8): every matching component is perfectly correlated ($|\text{corr}| = 1.000$), all cross-correlations are 0, and $|Z| = |Z_g|$ to machine precision. Because the two point clouds are identical up to reflecting individual axes, the clusters — and hence their number — must coincide. (In general the two pictures can differ slightly when one matrix is numerically better conditioned, but here the structure is clean enough that they agree exactly.)

10.2 What does cluster membership *mean* in each space?

Covariance space (3e). A vial's coordinates are the projections of its centred spectrum onto the top feature-space eigenvectors. Each coordinate says how strongly the vial expresses one dominant *pattern of spectral variation* (a weighted combination of frequencies). Two vials are close when they express these patterns to the same degree, i.e. their spectra are similar mixtures along the directions where samples differ most. Covariance space answers "*which mixture of frequencies does this vial express?*"

Gram space (3f). A vial's coordinates are the components of the top sample-space eigenvectors of G. Since G_{ij} is the similarity between vials i and j, each Gram eigenvector is a pattern over the 924 vials, and a vial's coordinate is its loading in that pattern — a compressed summary of how similar it is to every other vial. Gram space answers "*which other vials does this one resemble?*"

Both place vials with near-identical spectra at the same point, so proximity in either coordinate system means the underlying vials hold the same chemical concoction — which is why the two routes agree.

11 Part (h): how many unique food samples are there?

Finally we count the genuine concoctions. We use all 960 genuine vials — the 40 empty blanks removed, but the 36 high-norm vials *kept*, because part (h) tells us the samples are real fermented drinks, so that coherent, strongly-absorbing group is a genuine concentrated concoction, not an error. (This is exactly why

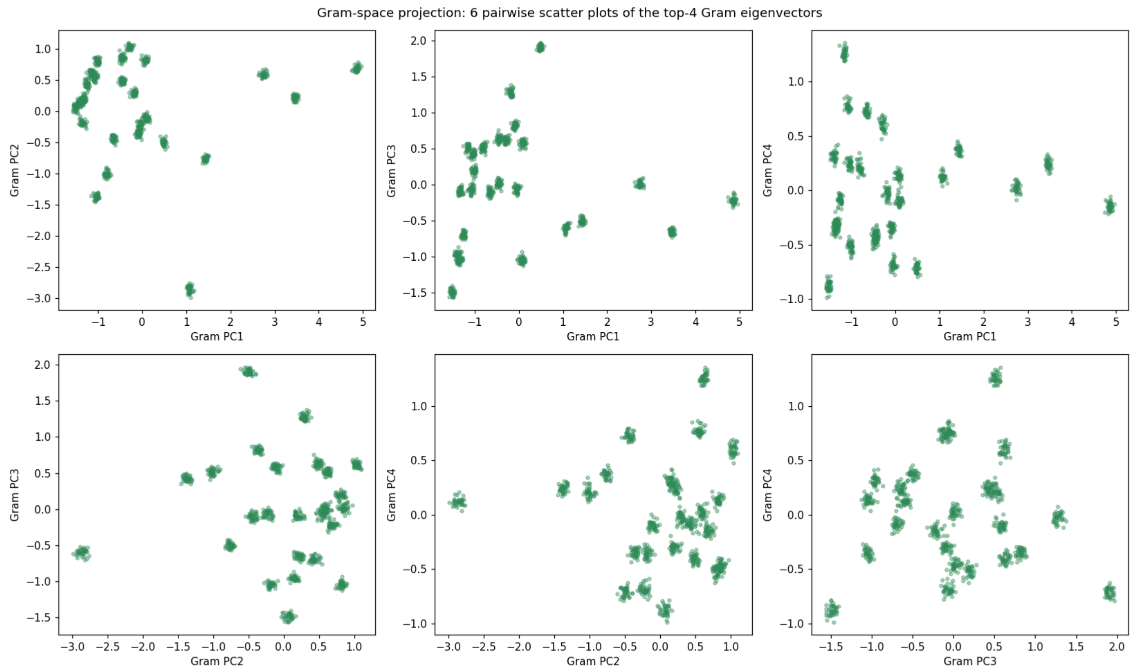

Figure 7: The six pairwise scatter plots of the top-4 Gram-space coordinates. They reproduce the covariance-space scatter of Figure 6 up to reflections, and again show ≈ 25 clusters.

the removal in §4 was kept reversible.) We cluster in the top-8 signal subspace (the elbow of §6), which denoises the data so that the groups separate cleanly.

Single-linkage connected components give 26 **clusters**, and the count is completely stable: it is 26 for every threshold ε from 0.25 to 0.40, and the leader clustering of §8–9 agrees. The decisive evidence is the *replicate structure* (Figure 9): the clusters have sizes $24 \times 37 + 2 \times 36 = 960$ — about 37 repeat measurements of each of 26 distinct samples. That clean, near-constant group size is the fingerprint of a designed experiment with 26 unique concoctions, comfortably below the "fewer than 100" stated in the prompt.

Remark 11.1. The covariance/Gram clusterings of §8–9 reported 25 because they were run on the 924-vial set that *excluded* the high-norm group. Restoring that one genuine concoction gives 26. This is the practical pay-off of having kept the cleaning decision reversible.

11.1 Which earlier parts were useful?

- **Part (a), cleaning.** Removing the 40 empties was essential: left in, they would form a spurious extra "cluster" of pure noise. Recognising the 36 high-norm vials as a *real* sample (and keeping the removal reversible) is what turns 25 into the correct 26.
- **Part (c), the elbow.** $k^\star = 8$ tells us the signal lives in ~ 8 dimensions, so we cluster in the top-8 principal components rather than the noisy 3401. This denoising is what makes the 26 groups cleanly separable; clustering the raw spectra would be swamped by noise. ($k^\star \approx 8$ also echoes the ~ 10 dominant molecules.)
- **Parts (e), (f), (g).** These supply the cluster count, and the fact that the covariance and Gram routes agree (g) shows the count is a real property of the data, not an artefact of one method.
- **Parts (b), (d).** These established that the data is low-dimensional and that the leading directions are chemically meaningful contrasts, which justifies the project-then-cluster pipeline.

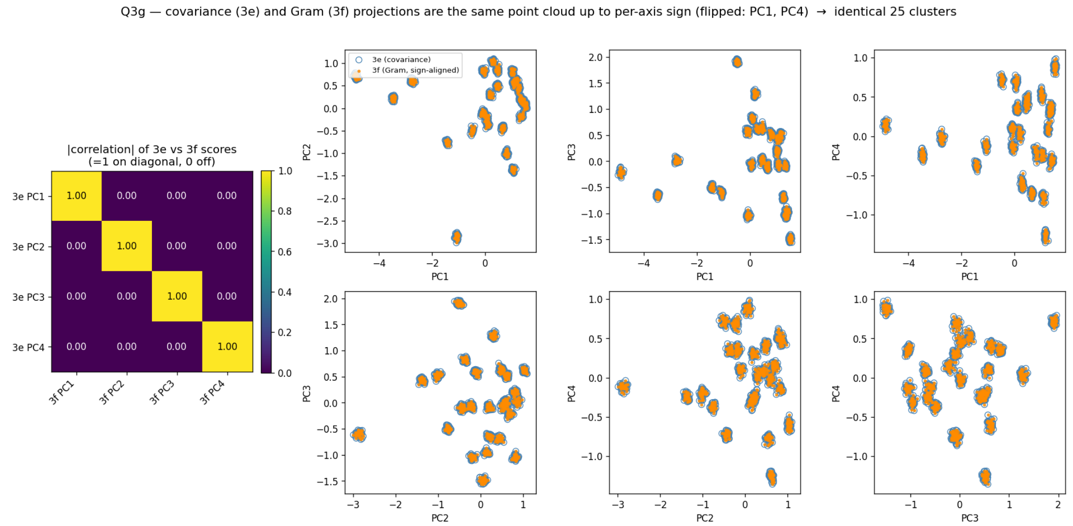

Figure 8: Comparison of the covariance-space (3e) and Gram-space (3f) coordinates: matching principal components are perfectly correlated ($|\text{corr}| = 1$) and distinct ones are uncorrelated, so the two scatter pictures are the same up to per-axis sign.

12 Conclusion

Starting from a raw 1000×3401 table, a short chain of linear-algebra tools recovered the hidden structure of the experiment. We removed 40 blank vials and 36 anomalous ones ($\widetilde{N} = 924$); found a total variance of $\mathrm{tr}(C) \approx 7.90$ concentrated in a few molecular bands; established by PCA that the data is strongly low-dimensional ($\lambda_1/\lambda_2 \approx 3.53$, an eight-dimensional signal subspace); interpreted the leading direction as a chemical contrast; and, by clustering the genuine vials two mathematically equivalent ways, determined that they comprise 26 **unique food samples**, each measured about 37 times.

The same principle threads through all three questions of the challenge: a high-dimensional system is governed by a small number of special directions. In Question 1 the dominant eigenvector was a Markov chain's long-time steady state; in Question 2 the Power Method found that direction by iteration, at a speed set by the spectral gap; and here in Question 3 the few leading eigenvectors of the covariance (equivalently the Gram) matrix compressed 3401-dimensional spectra into a picture clear enough to count the drinks.

A Glossary

Sample / feature
a sample is one vial (a row of X); a feature is one frequency (a column).
Norm $\|x\|$
the length $\sqrt{\sum_j x_j^2}$ of a vector; a one-number measure of size.
Transpose $X^\top$
the matrix with rows and columns swapped.
Mean-centering
subtracting each feature's average so every column averages zero.
Variance
mean squared deviation from the mean; how much a feature varies across samples.
Covariance matrix C
$\frac{1}{\widetilde{N}}\widetilde{X}^\top\widetilde{X}$ ($M \times M$, feature space); diagonal = variances.
Gram matrix G
$\frac{1}{\widetilde{N}}\widetilde{X}\widetilde{X}^\top$ ($\widetilde{N} \times \widetilde{N}$, sample space); G_{ij} = similarity of vials i, j.

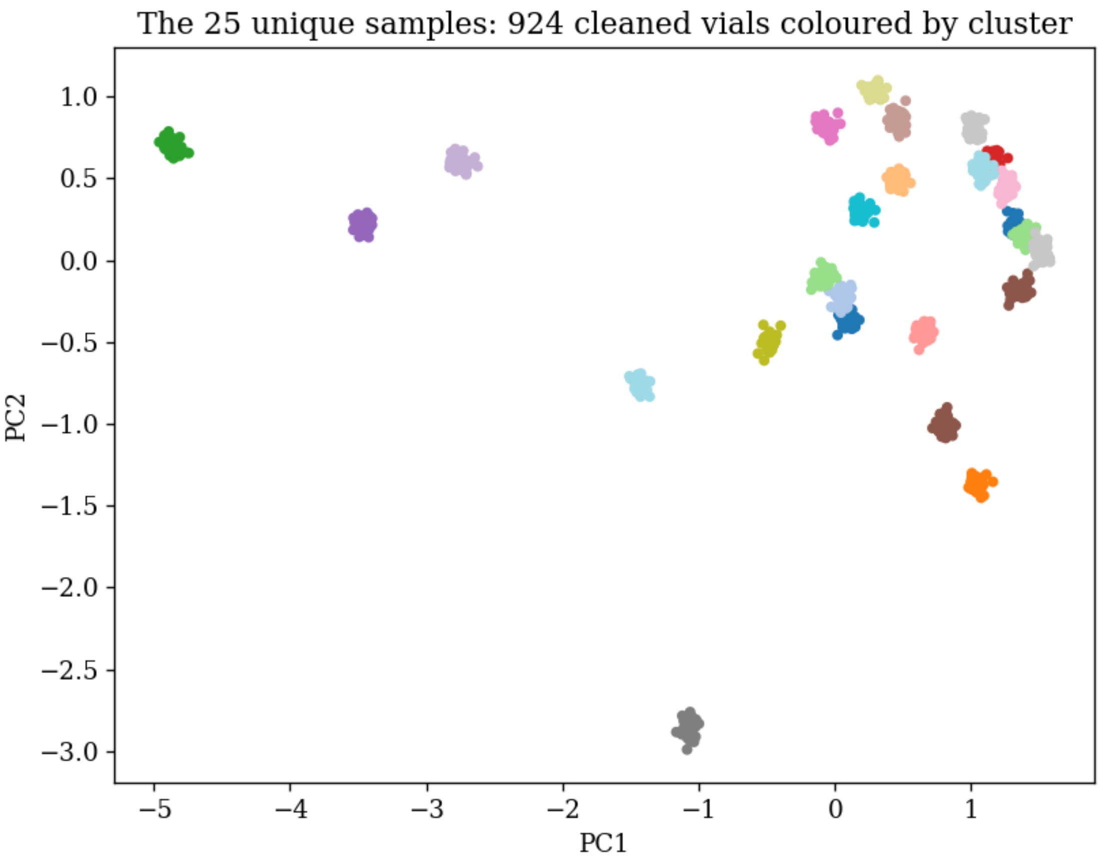

Figure 9: Cluster sizes for the 960 genuine vials in the top-8 subspace: 26 clusters of about 37 replicates each ($24 \times 37 + 2 \times 36 = 960$). The flat profile is the signature of a designed 26-sample experiment.

Trace
sum of the diagonal; for C it is the total variance $= \sum \lambda_i$.
Eigenvector / eigenvalue
a direction v with $Cv = \lambda v$; λ = variance along v.
Spectral theorem
a real symmetric matrix has real eigenvalues and orthonormal eigenvectors.
Positive semi-definite (PSD)
$w^\mathsf{T} C w \geq 0$ for all w; forces all $\lambda_i \geq 0$.
PCA
eigendecomposition of C; finds the directions of greatest variance.
SVD
the factorisation $\widetilde{X} = U \Sigma V^\mathsf{T}$; links C (V) and G (U) via σ_i.
Scree plot / elbow
sorted eigenvalues; the elbow $k^\star$ separates signal from the noise floor.
Spectral gap
$\lambda_1 - \lambda_2$ (or λ_1/λ_2); a wide gap = one dominant direction.
Score / projection
$z_i = \widetilde{x}_i \cdot \nu$, a sample's coordinate along an eigenvector.
Cosine similarity
$(a \cdot b)/(\|a\|\|b\|)$; scale-invariant comparison of direction.
Cluster
a tight, well-separated group of points; here, vials of one concoction.

B Summary of key numerical results

Quantity	Value
File shape (as stored)	$(3401, 1000)$; transposed, $M = 3401$ (text says 3041)
Empty vials removed (3a-i)	40 (rows 960–999)
High-norm vials removed (3a-ii)	36 (rows 924–959)
Cleaned samples $\widetilde{N}$	924
Exact-zero fraction (empty / real)	50.2% / 24.5%
Max-variance feature $j^\star$ (3b)	910 (variance ≈ 0.0199)
Total variance tr(C) (3b)	≈ 7.90
Dominance ratio λ_1/λ_2 (3c)	≈ 3.53
Elbow $k^\star$ (3c)	8 (largest drop $\lambda_8/\lambda_9 \approx 4.1$)
Components for 95% variance (3c)	667
ν_1 vs centred / raw / mean spectrum (3d)	cos ≈ 0.92 / 0.67 / 0.03
Clusters in covariance / Gram space (3e/3f)	25 / 25 (on the 924 set)
Cross-method correlation (3g)	$\lvert$corr$\rvert = 1$ matched, 0 otherwise
Unique food samples (3h)	**26** ($\approx$ 37 replicates each)